

Module 01

Ansible Introduction

DevOps end to end services & solutions

Agenda

- > What is Ansible?
- > Why Ansible?
- > Where is it used?
- > Ansible Architecture
- > Ansible vs Chef vs Puppet
- > Hands-On Lab-01

What is Ansible?

Ansible is an open-source IT automation tool.

- > Automates configuration management, application deployment, cloud provisioning, and orchestration.
- > Developed by Michael DeHaan, now maintained by Red Hat.
- > **Simple:** Uses human-readable YAML.
- > **Agentless:** No software needed on managed nodes.



Why Ansible?



Agentless

Uses SSH (or WinRM). No agents to install or update on managed nodes.



Simple

Written in YAML (Yet Another Markup Language), which is easy to read and write.



Idempotent

Ensures predictable results. Running the same playbook twice won't break anything.



Extensible

Supports custom modules, plugins, and dynamic inventory.

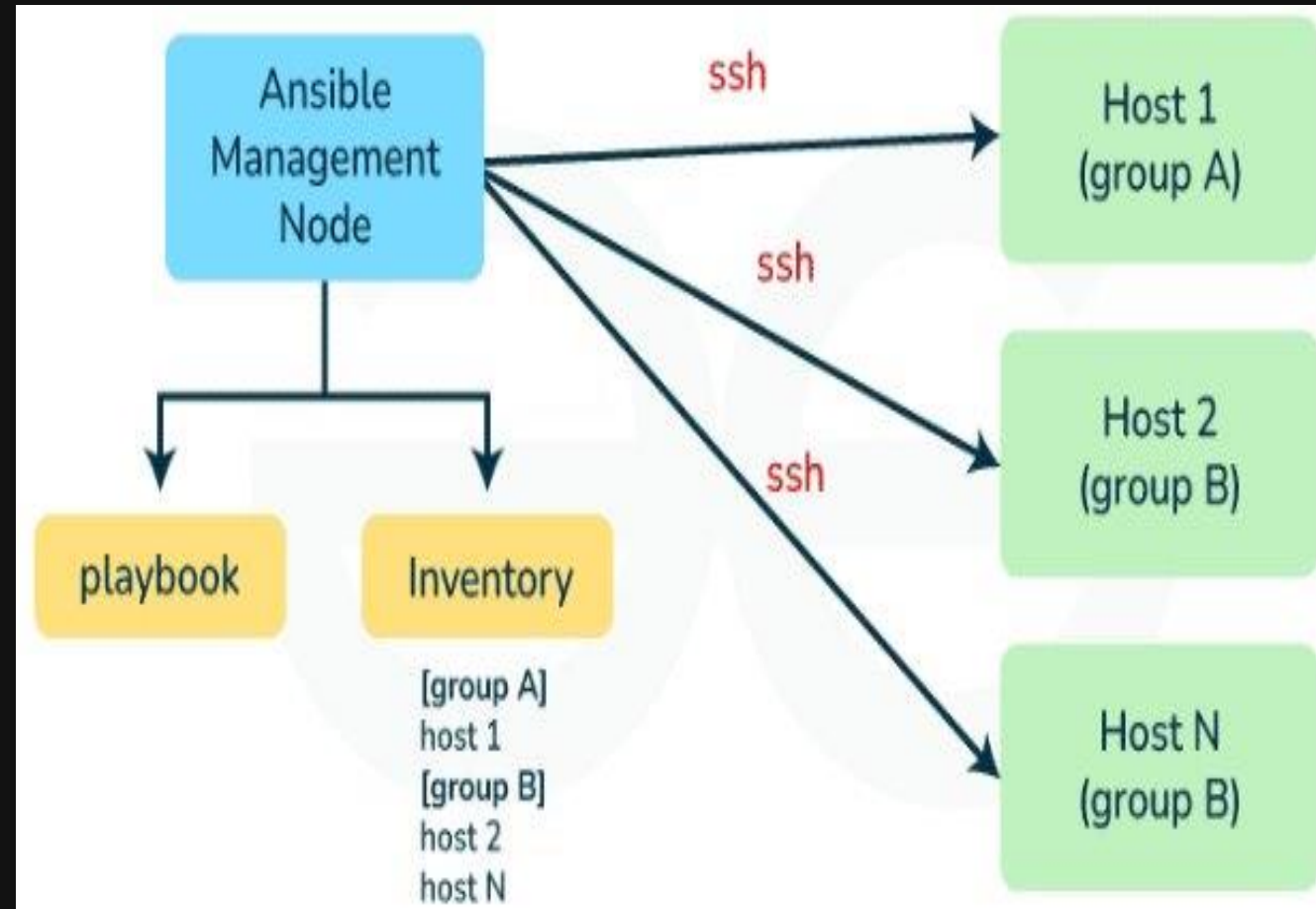
Where is it used?

- > **Configuration Management:** Managing Linux/Unix and Windows servers.
- > **Application Deployment:** Deploying applications across environments.
- > **Orchestration:** Coordinating multi-tier deployments.
- > **Cloud Provisioning:** Setting up infrastructure on AWS, Azure, GCP.
- > **Network Automation:** Managing network devices (Cisco, Juniper, etc.).

Ansible Architecture

Ansible is an open-source IT automation tool.

- > Manager Node – Control Node.
- > Agent – Managed Nodes.
- > **Inventory**
- > **Playbook**
- > Modules
- > Plugins



Ansible Architecture

Control Node

The machine where Ansible is installed.
Executes all playbooks. Requires Python + SSH.

Managed Nodes

The target machines (hosts) being configured. No Ansible agent is needed here.

Communication happens via SSH (Linux) or WinRM (Windows).

Core Components

Inventory

A file (INI, YAML, or dynamic script) that lists the managed hosts and organizes them into groups.

```
[webservers]
web1.example.com
web2.example.com

[dbservers]
db1.example.com
```

Modules

Units of work that Ansible pushes out to managed nodes.

- > Examples: copy, ping, service, yum, apt.
- > Modules are executed remotely and then removed.

Core Components

Playbooks

The instruction manuals for Ansible. Written in YAML.

- > Define tasks, roles, and variables.
- > Describe the desired state of the system.
- > Orchestrate steps across multiple machines.

Plugins

Code that plugs into the core Ansible engine to extend its functionality.

- > **Callback:** Hook into events (e.g., logging).
- > **Connection:** How to talk to hosts (SSH, Docker).
- > **Lookup:** Retrieve data from outside sources.

Ansible vs. Others



Ansible

Push-based
Agentless
YAML
(Simple)



Chef

Pull-based
Agent Required
Ruby
(Complex)



Puppet

Pull-based
Agent Required
Puppet DSL
(Declarative)

Ansible is often preferred for its simplicity and ease of setup.

Hands-On – Lab 01

Getting Started

In this lab, we will explore the basics of Ansible.

- > Set up a Control Node.
- > Configure an Inventory file.
- > Run ad-hoc commands using modules (ping, shell).
- > Verify connectivity to Managed Nodes.

Questions?

Thank you for participating.

Feel free to reach out:
<https://dinghy-e2e.com>

